

1. Write a program to check whether a number is even or odd using switch-case.

```
2. package Switch_case;
3.
4. import java.util.Scanner;
5.
6. public class Odd_even {
7.
8.     public static void main(String[] args) {
9.         // TODO Auto-generated method stub
10.        Scanner sc = new Scanner(System.in);
11.
12.        System.out.print("Enter a number: ");
13.        int num = sc.nextInt();
14.
15.        switch (num % 2) {
16.            case 0:
17.                System.out.println("Even");
18.                break;
19.            case 1:
20.            case -1:
21.                System.out.println("Odd");
22.                break;
23.        }
24.
25.        sc.close();
26.
27.    }
28.
29. }
```

Output:

Enter a number: -24

Even

2. Write a program that takes a character as input and checks whether it is a vowel or consonant using switch-case.

```
3. package Switch_case;
4.
5. import java.util.Scanner;
6.
7. public class Vowel_consonant {
8.
9.     public static void main(String[] args) {
10.        // TODO Auto-generated method stub
11.        Scanner sc = new Scanner(System.in);
12.
13.        System.out.print("Enter a character: ");
14.        char ch = Character.toLowerCase(sc.next().charAt(0));
15.
16.        switch (ch) {
17.            case 'a':
18.            case 'e':
19.            case 'i':
20.            case 'o':
21.            case 'u':
22.                System.out.println("Vowel");
23.                break;
24.            default:
25.                if (ch >= 'a' && ch <= 'z') {
26.                    System.out.println("Consonant");
27.                } else {
28.                    System.out.println("Not an alphabet");
29.                }
30.        }
31.    }
32. }
```

```

30.     }
31.
32.     sc.close();
33.
34. }
35.
36. }

```

Output:

Enter a character: T

Consonant

3. Write a program that takes a grade (A, B, C, D, F) and displays the performance message.

```

4. package Switch_case;
5.
6. import java.util.Scanner;
7.
8. public class Grade {
9.
10.     public static void main(String[] args) {
11.         // TODO Auto-generated method stub
12.         Scanner sc = new Scanner(System.in);
13.
14.         System.out.print("Enter grade: ");
15.         char grade = Character.toUpperCase(sc.next().charAt(0));
16.
17.         switch (grade) {
18.             case 'A':
19.                 System.out.println("Excellent");
20.                 break;
21.             case 'B':
22.                 System.out.println("Very Good");
23.                 break;
24.             case 'C':
25.                 System.out.println("Good");
26.                 break;
27.             case 'D':
28.                 System.out.println("Average");
29.                 break;
30.             case 'F':
31.                 System.out.println("Fail");
32.                 break;
33.             default:
34.                 System.out.println("Invalid Grade");
35.         }
36.
37.         sc.close();
38.
39.     }
40.
41. }

```

Output:

Enter grade: B

Very Good

4. Create a menu-driven program for ATM operations:

- 1 → Check Balance
- 2 → Deposit
- 3 → Withdraw
- 4 → Exit

Perform the operation based on user choice.

```
package Switch_case;

import java.util.Scanner;

public class ATM {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Scanner sc = new Scanner(System.in);

        double balance = 5000.00;
        int choice;
        boolean isRunning = true;

        System.out.println("ATM ---");

        while (isRunning) {
            System.out.println("Select Operation:");
            System.out.println("1: Check Balance");
            System.out.println("2: Deposit");
            System.out.println("3: Withdraw");
            System.out.println("4: Exit");
            System.out.print("Enter choice: ");

            choice = sc.nextInt();

            switch (choice) {
                case 1:
                    System.out.println("Current Balance: Rs." + balance);
                    break;

                case 2:
                    System.out.print("Enter deposit amount: ");
                    double deposit = sc.nextDouble();
                    if (deposit > 0) {
                        balance += deposit;
                        System.out.println("Deposited. New Balance: Rs." + balance);
                    } else {
                        System.out.println("Invalid amount.");
                    }
                    break;

                case 3:
                    System.out.print("Enter withdrawal amount: ");
                    double withdraw = sc.nextDouble();
                    if (withdraw > 0 && withdraw <= balance) {
                        balance -= withdraw;
                        System.out.println("Withdrawn. Remaining Balance: Rs." + balance);
                    } else if (withdraw > balance) {
                        System.out.println("Error: Insufficient amount.");
                    } else {
                        System.out.println("Error: Invalid amount.");
                    }
            }
        }
    }
}
```

```

        }
        break;

    case 4:
        System.out.println("Thank you.");
        isRunning = false;
        break;

    default:
        System.out.println("Invalid choice. Try again.");
    }
}
}
}
}

```

Output:

```

ATM ---
Select Operation:
1: Check Balance
2: Deposit
3: Withdraw
4: Exit
Enter choice: 3
Enter withdrawal amount: 200
Withdrawn. Remaining Balance: Rs.4800.0
Select Operation:
1: Check Balance
2: Deposit
3: Withdraw
4: Exit
Enter choice: 4
Thank you.

```

5. Create a program where user selects food item number and the program displays item name and price.

```

package Switch_case;

import java.util.Scanner;

public class Food_item {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.println("Food Menu -----");
        System.out.println("1: Burger");
        System.out.println("2: Pizza");
        System.out.println("3: Pasta");
        System.out.println("4: Sandwich");

        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();
    }
}

```

```

switch (choice) {
    case 1:
        System.out.println("Food: Burger");
        System.out.println("Price: ₹150");
        break;

    case 2:
        System.out.println("Food: Pizza");
        System.out.println("Price: ₹300");
        break;

    case 3:
        System.out.println("Food: Pasta");
        System.out.println("Price: ₹200");
        break;

    case 4:
        System.out.println("Food: Sandwich");
        System.out.println("Price: ₹100");
        break;

    default:
        System.out.println("Invalid choice!");
}

}

}

```

Output:

Food Menu -----

1: Burger

2: Pizza

3: Pasta

4: Sandwich

Enter your choice: 4

Food: Sandwich

Price: ₹100

6. Write a program to check whether a number is even or odd using switch-case.

Ans- SAME QUESTION AS NUMBER Q1.

7. Write a program that converts currency based on user choice (e.g., INR to USD, INR to EUR, etc.)

```
package Switch_case;
```

```
import java.util.Scanner;
```

```
public class Currency {
```

```

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
    }
}

```

```

System.out.println("Currency -----");
System.out.println("1: INR to USD");
System.out.println("2: INR to EUR");
System.out.println("3: INR to GBP");

System.out.print("Enter your choices from 1 to 3: ");
int choice = sc.nextInt();

System.out.print("Enter amount in INR: ");
double amount = sc.nextDouble();

double result;

switch (choice) {
    case 1:
        result = amount * 0.012;
        System.out.println("USD: " + result);
        break;

    case 2:
        result = amount * 0.011;
        System.out.println("EUR: " + result);
        break;

    case 3:
        result = amount * 0.0095;
        System.out.println("GBP: " + result);
        break;

    default:
        System.out.println("Invalid choice");
}
}
}

```

Output:

```

Currency -----
1: INR to USD
2: INR to EUR
3: INR to GBP
Enter your choices from 1 to 3: 2
Enter amount in INR: 100
EUR: 1.0999999999999999

```

8. Create a program that checks user role (Admin, User, Guest) and displays access permissions.

```

package Switch_case;

import java.util.Scanner;

public class User_role {

    public static void main(String[] args) {

```

```

        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter role (Admin/User/Guest): ");
        String role = sc.nextLine();

        if (role.equalsIgnoreCase("Admin")) {
            System.out.println("Access: Full access to system");
        } else if (role.equalsIgnoreCase("User")) {
            System.out.println("Access: Limited access");
        } else if (role.equalsIgnoreCase("Guest")) {
            System.out.println("Access: View only");
        } else {
            System.out.println("Invalid role");
        }
    }
}

```

Output:

```

Enter role (Admin/User/Guest): User
Access: Limited access

```

9. Write a program to calculate the factorial of a given number using a while loop.

```

package Control_Statements;

import java.util.Scanner;

// Factorial

public class Factorial {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");
        int a = sc.nextInt();

        int fact = 1;

        while(a>0) {
            fact *= a;
            a--;
        }

        System.out.print("Factorial is: "+fact);

        sc.close();
    }
}

```

Output:

```

Enter number: 5
Factorial is: 120

```

10. Write a program to print the multiplication table of a given number using a while loop.

```
package Control_Statements;

import java.util.Scanner;

public class Table_multi {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int i = 1;

        while (i <= 10) {
            System.out.println(num + " x " + i + " = " + (num * i));
            i++;
        }
    }
}
```

Output:

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

11. Write a program to count the number of digits in a given number using a while loop.

```
package Control_Statements;

import java.util.Scanner;

public class NUm_of_dig {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        int c=0;

        while (num>0) {
            num /=10;
        }
    }
}
```



```

        c++;
    }
    System.out.println("Number of digits: "+c);

}

}

```

Output:

Enter a number: 0123456789

Number of digits: 9

12. Write a program to reverse a given number using a while loop.

```

package Control_Statements;

import java.util.Scanner;

public class Reverse_num {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt();
        int reverse = 0;

        while (num != 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num /= 10;
        }

        System.out.println("Reversed Number: " + reverse);

    }

}

```

Output:

Enter number: 546

Reversed Number: 645

13. Write a program to check whether a number is a palindrome using a while loop.

```

package Control_Statements;

import java.util.Scanner;

public class Palindrome {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

```

```

System.out.print("Enter number: ");
int num = sc.nextInt();
int temp = num;
int reverse = 0;

while (num > 0) {
    int digit = num % 10;
    reverse = reverse * 10 + digit;
    num /= 10;
}

if (temp == reverse) {
    System.out.println(temp + " is a Palindrome.");
} else {
    System.out.println(temp + " is not a Palindrome.");
}

}

}

```

Output:

Enter number: 121
121 is a Palindrome.

14. Write a program to find the sum of digits of a number using a while loop.

```

package Control_Statements;

import java.util.Scanner;

public class Dig_sum {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        int sum = 0;

        while (num > 0) {
            sum += (num % 10);
            num /= 10;
        }

        System.out.println("Sum of digits: " + sum);

    }

}

```

Output:

Enter a number: 1234
Sum of digits: 10

15. Write a program to check whether a number is an Armstrong number using a while loop.

```
package Numbers;

import java.util.Scanner;

public class Armstrong {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num = sc.nextInt();
        int temp = num;
        int result = 0;

        while (num > 0) {
            int digit = num % 10;
            result += (digit * digit * digit);
            num /= 10;
        }

        if (result == temp) {
            System.out.println(temp + " is an Armstrong number.");
        } else {
            System.out.println(temp + " is not an Armstrong number.");
        }
    }
}
```

Output:

```
Enter number: 625
625 is not an Armstrong number.
```

16. Write a program to check whether a number is prime using a while loop.

```
package Numbers;

import java.util.Scanner;

public class Is_prime {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");
        int num = sc.nextInt();

        boolean isPrime = true;

        if(num <= 1) {
            isPrime = false;
        }
        else {
            for(int i=2; i<Math.sqrt(num); i++) {
```

```

        if(num%i == 0)
        {
            isPrime = false;
            break;
        }
    }
}
if(isPrime) {
    System.out.print(num+" is Prime number");
}
else {
    System.out.print(num+" is not prime number");
}
}
}

```

Output:

Enter number: 17
17 is Prime number

17. Write a program to print Fibonacci series up to N terms using a while loop.

```

package Control_Statements;

import java.util.Scanner;

public class Fibonacci {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter terms: ");
        int n = sc.nextInt();

        int first = 0;
        int second = 1;
        int i = 1;

        System.out.println("Fibonacci Series up to " + n + " terms:");

        while (i <= n) {
            System.out.print(first + " ");

            int temp = first + second;
            first = second;
            second = temp;
            i++;
        }
    }
}

```

Output:

Enter terms: 6

Fibonacci Series up to 6 terms:

0 1 1 2 3 5

